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Beacon correction method for inter-satellite laser communication

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ABSTRACT

Positioning errors are caused by wavefront distortion owing to the thermal deformation of optical systems. To address this issue, we propose a restoration algorithm to correct for the beacon centroid, which is important for inter-satellite laser communication links. Based on the far-field distribution model of the distortion facula, the objective function is established to solve the wavefront distortion for reducing the pointing error of the optical communication system. The feasibility of this approach is verified by theoretical analysis and experimental verification, including in-orbit experimental verification. The proposed approach is likely to be useful for designing inter-satellite laser communication systems.

1. Introduction

To establish a laser communication link between two satellites separated by thousands of kilometers, communication terminals should be able to quickly acquire and stably track communication beams. For this, communication terminals should have high detection sensitivity and strong communication capabilities. During the establishment of the laser communication link, the pointing, acquiring, and tracking (PAT) process is crucial, and a charge coupled device (CCD) detector is the key component of any PAT system [1,2]. For successful tracking, the positioning accuracy of the beacon directly affects the accuracy of the target tracking, thus affecting the communication quality [3]. When a laser communication link is established between satellites, the far field of the received ideal beam can be approximated as a plane wave. However, owing to the spacecraft operating in the outer space, its working state is affected by the space thermal environment, which is strongly determined by the sun. The temperature conditions of the optical communication terminals are complex. The harsh space environment conditions include rapid changes in high and low temperatures, and cosmic radiation, and control precision of the thermal system of an optical communication system may also influence the positioning precision of the optical communication system [4,5]. In addition, the method is introduced for studying the effects of thermal deformations on optical systems for space application [6].

In the inter-satellite laser communication process, to realize the acquisition of signals and tracking of satellites, it is necessary to accurately and quickly locate the facula received on the CCD detector to obtain the position of the communication satellite, and to achieve fast acquisition and stable tracking by adjusting the attitude of the satellite. Among the many positioning algorithms, the most frequently used gray-scale centroid localization algorithm is simple, which meets real-time requirements and yields relatively high positioning accuracy for satellite laser communication. However, the gray centroid algorithm also has certain limitations. Only when the energy distribution of the facula is relatively uniform, the gray centroid algorithm can yield better positioning. However, owing to the influence of aberrations caused by the thermal deformation of the optical system, especially the existence of tilt and coma, the energy distribution of the facula becomes uneven, so that the locating process using the gray centroid algorithm yields large positioning errors, which will inevitably affect the acquiring and tracking beacon.

These unfavorable factors unpredictably change the basic parameters of the mirror and inevitably affect the accuracy of tracking, thus affecting the quality of communication. Because wavefront distortions caused by changes in the optical system's surface shape affect the transmission of optical signals, the light field received by the system's CCD detector is distorted, causing the positioning algorithm to generate positioning errors, which is turns affects the acquisition and tracking. At present, much research has been done on the influence of wavelength-distorted beams on positioning errors, but research on the beacon distortion correction considering thermal deformation-related wavefront distortion is very scarce. Thus, in the present work, we addressed the question of beacon distortion correction, for improving the reliability and robustness of communication systems. In addition, a verification experiment simulates the in-orbit case, which gives a new thought for dealing with positioning errors caused by wavefront distortion owing to the thermal deformation of optical systems. As the number of years in-orbit increases, the performance of thermal control system will degrade. Decline of temperature control accuracy will inevitably result in thermal inhomogeneity of the laser communication terminal. The approach has practical application meaningful. By the approach, the hardware complexity is not increased. This method realized by the parallel processing system is relatively easy and effective.

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The theoretical analysis is introduced in Section 2. The simulation is given in Section 3 and the verification experiment is conducted in Section 4, and the Conclusions section summarizes our findings.

2. Theoretical analysis

To establish the far-field distribution model of the wavefront-distorted beam, the cause of the beam distortion is first analyzed, which corresponds to the thermal deformation of the main mirror of the Cassegrain optical antenna. When a beam passes through an optical system that undergoes thermal deformation, the beam is distorted. The influence of the wavefront phase distortion introduced by the optical system on the transmitted light beam can be captured by multiplying the original optical intensity distribution function by a distortion phase factor $\exp[j\Phi(x,y)]$ [7]. For inter-satellite laser communication systems, because the distance between the two satellites is very large, the light field at the receiving terminal can be approximately regarded as parallel light without distortion, and the distorted parallel beam can be calculated by [8]

$$E(x,y) = \mathrm{C}\cdot\exp(j\Phi(x,y)) \tag{1}$$

where C is a constant.

The phase of the wavefront distortion can be described by the Zernike polynomial as

$$\Phi(x,y) = \sum a_n Z_n \tag{2}$$

where Z_{n} is the Zernike polynomial and a_{n} is the Zernike polynomial coefficient.

The communication distance between satellites is very long, and the transmitted beams satisfy the Fraunhofer diffraction requirements. The complex amplitude distribution of the receiving plane light field is [8]

$$\mathrm{U}(x,y) = \mathrm{C} * \iint \exp[j\Phi(x,y)] * \exp\left[-j\frac{2\pi}{\lambda z}(xx_0+yy_0)\right]dx_0dy_0 \tag{3}$$

The distribution of the light intensity of the receiving plane is the square of the complex amplitude distribution of the receiving plane light field. The far-field distribution model of the distortion facula, under the influence of the beam wavefront distortion, that is

$$\mathrm{I}(x,y) = U(x,y)\cdot U^*(x,y) = |U(x,y)|^2 \tag{4}$$

In the inter-satellite laser communication, a beacon can be collected on a CCD, and the wavefront distortion phase of the transmitted beam is reconstructed based on the beacon detection.

At this time, the process of the wavefront phase reconstruction can be regarded as a nonlinear programming problem. The Cassegrain optical system is affected by thermal deformation, which adds a distorted phase to the light field. The distortion phase affects the energy distribution of the facula, and the facula on the CCD detector exhibits various aberrations. In the above analysis, the Zernike polynomials can be used to describe these aberrations. The phase distribution of the distortion can be obtained as a combination of the Zernike polynomials. If the Zernike coefficient in the polynomial combination can be obtained, the phase of the distortion can be described. The wavefront distribution function of the beam is obtained. The combination of the Zernike polynomials is included in the expression of the optical intensity distribution.

An objective function is established between the in-orbit measurement phase and the calibration phase. The calibration phase can be obtained by laboratory measurement when the optical terminal is placed on a vibration isolation table in a vacuum. In this condition, the facula is not distorted, the Zernike polynomial and the Zernike polynomial coefficient can be computed based on the calibration phase. The calibration results can be considered as a standard for compensating the system error caused by the wavefront distortion in orbit.

And the formulas for the Fourier transform are used for computing the on-orbit measurement phase and the calibration phase, these are

$$\Phi_{1,i}(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} U_{1,i}(x,y)\exp[-j2\pi(ux+vy)]dxdy \tag{5}$$

$$\Phi_{2,i}(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} U_{2,i}(x,y)\exp[-j2\pi(ux+vy)]dxdy \tag{6}$$

$$U_{1,i}(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \Phi_{1,i}(u,v)\exp[j2\pi(ux+vy)]dudv \tag{7}$$

$$U_{2,i}(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \Phi_{2,i}(u,v)\exp[j2\pi(ux+vy)]dudv \tag{8}$$

where $U_{1,i}(x,y)$ is the gray distribution of the in-orbit collected image, $U_{2,i}(x,y)$ is the calibration gray distribution, $\Phi_{1,i}(u,v)$ is phase of the image collected by the CCD detector at the ith instance, and $\Phi_{2,i}(u,v)$ is the calibration phase of the undistorted facula at the ith instance. The number of instances is $i = 5$; that is, five frame images are concentrated when solving a set of the Zernike polynomial coefficients.

The objective function is

$$f = \sum_i \left|\Phi_{1,i}(u,v) - \Phi_{2,i}(u,v)\right|^2 = \sum_i \left|\sum a_n Z_n - \sum b_n Z_n\right|^2 \tag{9}$$

where a_n is the Zernike polynomial coefficient when phase of the image is $\Phi_{1,i}(u,v)$, and b_n is the Zernike polynomial coefficient when phase of the image is $\Phi_{2,i}(u,v)$.

The objective function is considered as a standard for evaluating the distortion. If the facula is undistorted, the actual measured result is the most consistent with the calibration result, and the objective function is very small. However, beam wavefront distortion happens, and the objective function will get bigger. We need to compensate the distortion according to the objective function by reconstructing the distorted light field. When the objective function attains a minimum, the reconstruction result is the most consistent with the calibration result. The correctional coefficient of the Zernike polynomial at this time is obtained.

The minimum value of this objective function can be performed using the conjugate gradient method. The conjugate gradient method is an unconstrained optimization algorithm; in terms of its complexity, it occupies a space between the Newton method and the steepest descent method. When solving unconstrained linear optimization problems, only the objective function itself and its first derivative are needed, which significantly reduces the computational and memory burden associated with the calculation of the second derivative. The conjugate gradient method is a simple, efficient, and widely used method for unconstrained linear optimization problems [9].

The specific flow of conjugate gradient method is as follows:

Step 1: Set the error $\varepsilon = e^{-4}$ and starting point x_0, and calculate $g_0 = \nabla f(x_0)$, let the maximum number of iterations $k < 5000$,

Step 2: If $\|g_k\| \le \varepsilon$, so stop the calculation and output $x^* \approx x_k$,

Step 3: Calculate the search direction d_k,

$$d_k = \begin{cases} -g_k (k=0) \\ -g_k + \beta_{k-1}d_{k-1} (k \ge 1) \end{cases}$$

where $k \ge 1$, $\beta_{k-1} = \frac{g_k^T g_k}{g_{k-1}^T g_{k-1}}$,

Step 4: Determine the search step size α_k using the exact line search method,

Step 5: Order $x_{k+1} = x_k + \alpha_k d_k$, and calculate $g_{k+1} = \nabla f(x_{k+1})$,

Step 6: Order $k = k+1$, go to step 2.

When using the conjugate gradient algorithm described above to solve the minimum value of the objective function, the first derivative of the objective function is needed, and the partial derivative of the objective function is obtained by using each Zernike polynomial coefficient to form a matrix.

Finally, by Eqs. (7) and (8), the correctional gray distribution can be precisely computed, and the facula correction position can be obtained accurately by using the gray centroid algorithm.

In the absence of distortion, it is assumed that the facula grayscale image is composed of (M×N) pixels, the coordinates of each pixel in the image are (x, y), the gray value of each pixel is $g(x, y)$, and the centroid coordinates are (x_C, y_C). According to the centroid principle, the centroid coordinates are

$$x_c = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} x \times g(x,y)}{\sum_{i=1}^{M}\sum_{j=1}^{N} g(x,y)}, y_c = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} y \times g(x,y)}{\sum_{i=1}^{M}\sum_{j=1}^{N} g(x,y)} \tag{10}$$

When a distortion occurs, the intensity value of each pixel received on the receiving plane changes, and the gray value of each pixel in the image changes. The gray value of each pixel in the image after distortion is $g'(x, y)$, and the centroid of the distorted facula is (x'_c, y'_c):

$$x'_c = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} x \times g'(x,y)}{\sum_{i=1}^{M}\sum_{j=1}^{N} g'(x,y)}, y'_c = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} y \times g'(x,y)}{\sum_{i=1}^{M}\sum_{j=1}^{N} g'(x,y)} \tag{11}$$

The complex amplitude of the wavefront-distorted light field is

$$U'(x,y) = C \iint \exp(j\Phi(x,y)) \exp(-j\frac{2\pi}{\lambda z}(xx_0 + yy_0)) dx_0 dy_0 \tag{12}$$

Simplifying the above expression and taking the first-order approximation $\exp(j\Phi(x,y)) \approx 1 + j\Phi(x,y)$, the optical intensity distribution for the distorted facula becomes

$$I'(x,y) = \Big| \mathrm{C} \iint \exp(-j\frac{2\pi}{\lambda z}(xx_0 + yy_0)) dx_0 dy_0 + j\mathrm{C} \iint \Phi(x,y) \exp(-j\frac{2\pi}{\lambda z}(xx_0 + yy_0)) dx_0 dy_0 \Big|^2 \tag{13}$$

Based on Eq. (13), it can be found that the optical intensity distribution of the distorted facula contains two parts. The first part in the expression represents the optical intensity distribution of the facula when the optical system is not distorted. The second part represents the distribution of the light field caused by the wavefront distortion.

The reconstruction algorithm is summarized in Fig. 1. The objective function is established between the detected image and the calibration data, with the Zernike polynomial coefficient as the variable of the objective function. The initial value of each coefficient is in the -2 to 2 range, and the step size is 0.01. Based on the initial value, the conjugate gradient method is used to solve the Zernike polynomial coefficient when the objective function attains a minimum. Based on the Zernike polynomial coefficient, the relationship between the centroid coordinates before and after the facula distortion is used to solve the correction centroid coordinates. In the last step, the smallest of the stored restoration errors is determined, and the corresponding centroid coordinates are output, completing the algorithm flow.

The degree of distortion can be reflected by the state of the facula on CMOS camera, which is also the basis of the method in this paper. In this approach, many influencing factors are reflected in the state of the facula on CMOS camera. By this approach, these influencing factors on the facula can be taken into account. This method is real-time correction, not post-processing.

3. In-silico validation

Based on the distortion field facula far-field distribution model established in simulations, the restoration effect of the algorithm for the distortion facula was validated. The obtained distortion facula was used to validate the restoration effect of the algorithm proposed in this paper.

A FPGA (Field Programmable Gate Array) system was used to set the Zernike coefficient in the distortion facula model to obtain different distortion faculae. In the process of restoring the facula using the restoration algorithm, the calculated Zernike polynomial coefficients and the restored centroid coordinates were extracted and calibrated. The restoration effect of the restoration algorithm was analyzed using the conjugate gradient method. The faculae images are shown in Fig. 2.

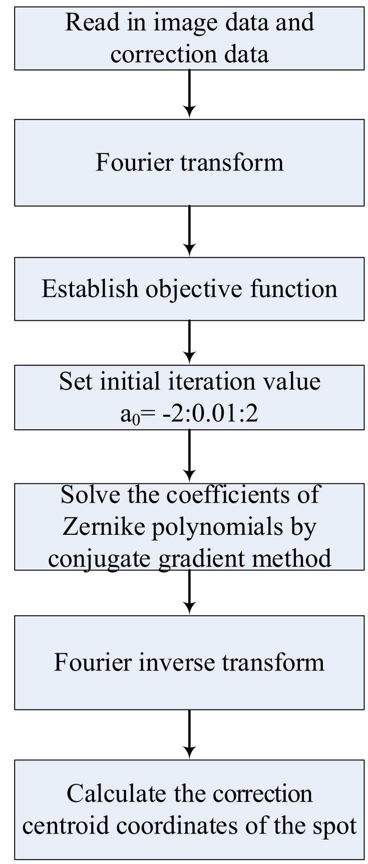


Fig. 1. Recovery algorithm flow chart.

Table 1
Recovery algorithm calculation data.

Calibration coordinates	Distortion facula coordinates	Correction coordinates by the new algorithm
(100.5,100.5)	(111.23,111.23)	(100.71,100.59)
(100.5,100.5)	(111.27,108.19)	(99.48,100.38)
(100.5,100.5)	(108.19,111.27)	(100.38,99.48)
(100.5,100.5)	(112.69,108.11)	(102.06,100.85)
(100.5,100.5)	(113.69,95.23)	(100.85,98.90)
(100.5,100.5)	(97.95,113.36)	(101.37,100.34)
(100.5,100.5)	(113.36,97.95)	(100.34,101.37)
(100.5,100.5)	(113.14,90.37)	(101.60,99.68)
(100.5,100.5)	(90.37,113.14)	(99.68,101.60)

Distorted faculae in Fig. 2 show that the deflection angles of the distorted faculae are different; thus, a reference value should be provided when using distorted faculae to validate the algorithm. The first four Zernike polynomial coefficients of Fig. 2 are that (a) 2,1,1,1 (b) 1,2,1,1(c) 1,1,2,1 (d) 1,1,1,2 (e) -2,1,1,1 (f) 1,-2,1,1 (g) 1,1,-2,1 (h) 1,1,1,-2, respectively.

During the restoration process, for a better recovery effect, the initial Zernike polynomial coefficients were set manually, and the calculated Zernike polynomial coefficients and the restored centroid positions were compared with the set values, as shown in Table 1.

Table 1 shows that the algorithm can achieve a better recovery effect. However, in the process of calculation, it is found that the Zernike polynomial coefficient is solved using the conjugate gradient method, and the value of the obtained coefficient is closely related to the initial coefficient value, especially when the set coefficient is negative. Moreover, the solution process for restoring the position of the centroid of the facula is also very dependent on the Zernike polynomial coefficient of the solution in the restoration algorithm.

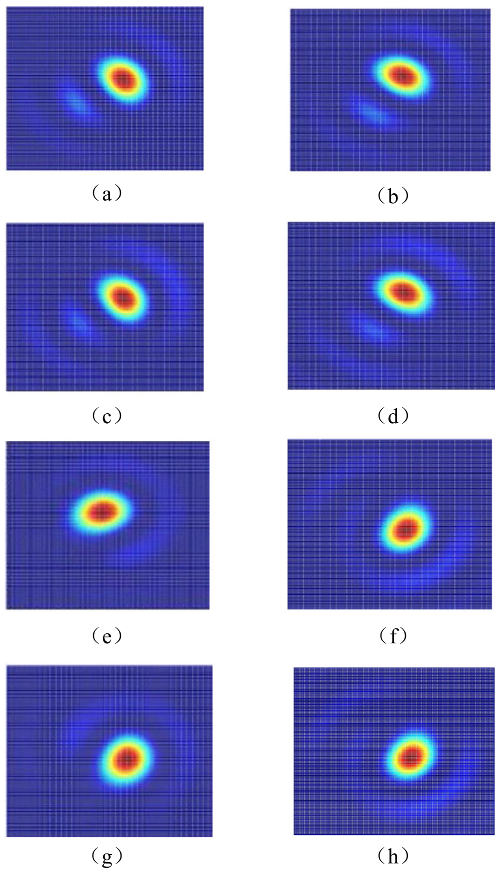


Fig. 2. Distortion faculae images.

To describe the restoration effect of the algorithm, the restoration error is defined in Eq. (14), and the restoration percentage is expressed in Eq. (15), where the centroid coordinates are (x_0, y_0), the coordinates of the centroid of the distortion facula are (x_1, y_1), and the coordinates of the centroid of the facula after restoration are (x_2, y_2).

$$\Delta x = |x_2 - x_0| \tag{14}$$

$$\eta = 1 - \frac{|x_2 - x_0|}{|x_1 - x_0|} \tag{15}$$

To more intuitively study the restoration effect of the restoration algorithm, according to the data in Table 1, the restoration error of the distortion plane in the X and Y directions is calculated, and the curve is shown in Fig. 3.

Fig. 3 shows that the algorithm can yield good restoration based on the Zernike polynomial coefficients, and visual inspection shows that the restoration error in both the X and Y directions is 1.6 pixels, while the average restoration errors calculated for the both directions is 0.72 pixels and 0.60 pixels, respectively. According to Eq. (15), the overall average error is 1.28 pixels, which corresponds to the recovery of 89.5%.

4. Experimental verification

An experiment was performed to validate the restoration algorithm. The terminal was a Cassegrain telescope with the receiving aperture of 250 mm. The beam was focused on a complementary metal oxide semiconductor (CMOS) camera after being transmitted through a reflector, an attenuator and a filter, and a computer controlled the CMOS camera to record images at 1000 frames/s, as shown in Fig. 4. Two sets of heaters were configured in a primary mirror and audition,

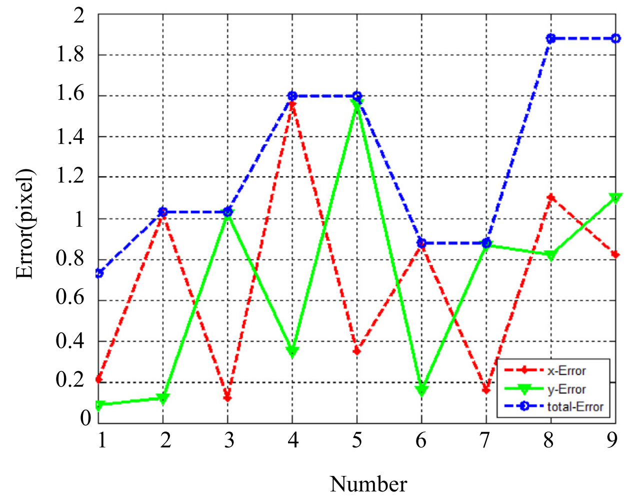


Fig. 3. Restoration error of the restoration algorithm in the X and Y directions.

Table 2
Parameters of the experimental system.

Number	Device	Parameters
1	CMOS Camera	Pixel size:4.8 μm, response wavelength:0.4–1 μm, window: 1280 × 1024 *(pixels)*
2	Aperture:	150 mm
3	Focal length:	600 mm
4	Lens	Effective aperture:5 mm,focal length: f=25 mm
5	Enlargement factor:	11

respectively. These heaters were used for producing mirror thermal deformation and for simulating the wavefront distortion from the thermal deformation of the optical system. In this experiment, the heaters were used with thermistors. Thermistors can feed back the variation of the temperature.

The system was placed on a vibration isolation table in a vacuum. The effect of vibration and atmospheric turbulence on the experimental results were excluded. The parameters of the experimental system is described in Table 2.

This is a dynamic regulation for the temperatures of heater I and heater II, and the collecting images of the CMOS camera are observed by us. When the Facula is disturbed, the temperatures of heater I and heater II are recorded.

The experiment is the approach of the ground equivalence verification for the design of the laser communication terminal. The location of Heat I and heater II are also based on the thermal control of the on-board terminal (HY-2 Satellite), and the thermal control data from the satellite are the setting standard for the heater I and heater II. The temperature variation ranges of heater I and heater II are based on in-orbit test data from the laser communication in-orbit experiment (HY-2 Satellite).

The undistorted facula image for this experiment is shown in Fig. 5 (The condition is that the temperature of heater I is 15.3 °C, and temperature of heater II is 22.4 °C), and some of the distortion faculae images collected by the CMOS camera are shown in Fig. 6. Thermal deformation resulted in the beacon distortion. The temperature of heater I is 35.4 °C, and temperature of heater II is 16.2 °C, so in this condition, the facula is shown in Fig. 6(a). The temperature of heater I is 42.6°C, and temperature of heater II is 17.3 °C, so in this condition, the facula is shown in Fig. 6(b).

The coordinates of the centroid of the undistorted facula were (20.5, 20.5), and were used as a standard for evaluating the performance of this algorithm. The maximum variation of coordinates of the centroid of

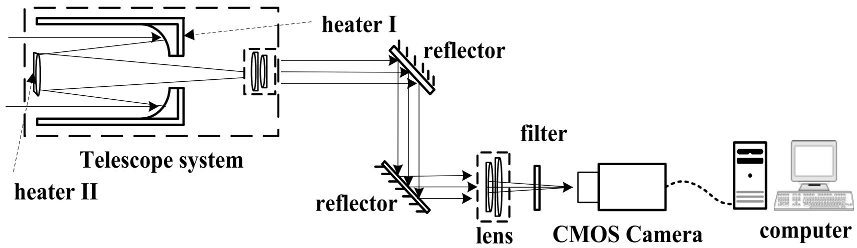


Fig. 4. Configuration of the experimental setup.

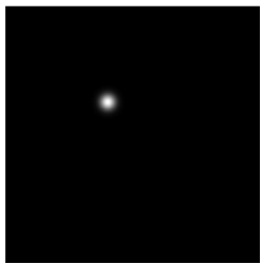
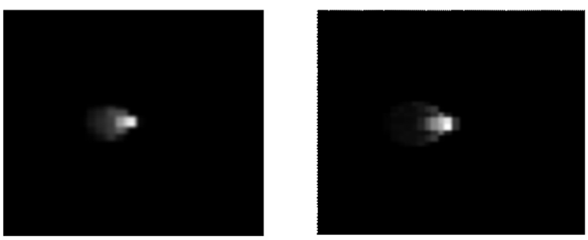


Fig. 5. Undistorted facula.



(a) (b)

Fig. 6. Distortion faculae images.

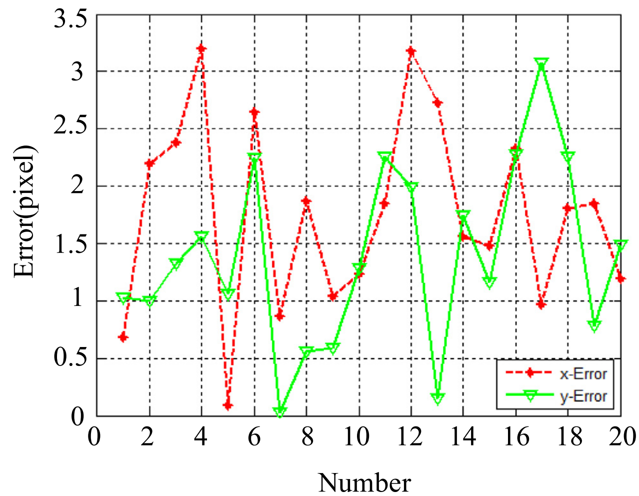


Fig. 7. Facula recovery error under specular distortion.

the undistorted facula was very small (<0.2 pixel), because the system was placed on a vibration isolation table in a vacuum.

The experimental results are shown in Table 3. Fig. 7 shows the correction results obtained using this algorithm. The red curve in Fig. 7 represents the restoration error of the restoration algorithm in the X direction, and the green curve represents the error distribution in the Y direction. The combined error for the two directions is determined by summing the restoration errors in the X and Y directions. The error distribution does not exhibit regularities. In this experimental system, 1pixel of CMOS camera is equal to 1 μrad.

The algorithm's restoration error for the distortion facula is within 3.5 pixels. The average is 1.76 pixels for the X direction, and the fractional recovery is 78.8%. The average is 1.39 pixels for the Y direction, and the fractional recovery is 65.2%. The total error of the facula restoration algorithm is within 4 pixels, the average restoration error is 2.36 pixels, and the average recovery is 69.4%. On average, the restoration algorithm yields good recovery and can reduce the positioning error to a certain extent, which is very important for improving the positioning accuracy and tracking accuracy of satellite communication systems.

In addition, the approach was also verified by the laser communication in-orbit experiment (HY-2 Satellite, China), which could give a new thought for dealing with positioning errors caused by wavefront distortion owing to the thermal deformation of optical systems. The maximum in-orbit recovery error is 3.5μrad, which is consistent with the ground test results. The approach is benefit for improving the reliability and robustness of the in-orbit optical communication system, too.

Table 3
Comparison between the correction coordinates and the calibration coordinates.

Calibration coordinates	Distortion facula coordinates	Correction coordinates by the new algorithm
(20.5,20.5)	(18.74,30.05)	(19.85,21.56)
(20.5,20.5)	(17.85,11.67)	(22.73,19.51)
(20.5,20.5)	(26.91,18.55)	(18.10,19.15)
(20.5,20.5)	(13.05,21.51)	(23.75,22.06)
(20.5,20.5)	(12.60,22.51)	(20.41,21.65)
(20.5,20.5)	(11.62,22.92)	(23.13,18.25)
(20.5,20.5)	(26.54,21.35)	(21.32,20.41)
(20.5,20.5)	(32.83,25.94)	(18.70,21.06)
(20.5,20.5)	(29.83,15.92)	(21.55,21.12)
(20.5,20.5)	(18.34,17.95)	(21.75,19.25)
(20.5,20.5)	(17.26,27.33)	(18.72,22.75)
(20.5,20.5)	(25.83,25.36)	(17.33,22.53)
(20.5,20.5)	(16.32,18.94)	(23.25,20.25)
(20.5,20.5)	(15.83,24.44)	(18.98,18.75)
(20.5,20.5)	(23.32,29.94)	(21.98,21.75)
(20.5,20.5)	(16.83,27.23)	(22.75,18.25)
(20.5,20.5)	(24.83,15.98)	(19.52,23.55)
(20.5,20.5)	(25.21,25.45)	(22.32,22.75)
(20.5,20.5)	(17.83,30.44)	(18.65,21.25)
(20.5,20.5)	(25.27,18.66)	(21.75,22.03)

5. Conclusions

In this paper, the problem of the facula positional error caused by thermal deformation was addressed. The Zernike polynomial was used to describe the wavefront distortion phase. In the far-field Fraunhofer diffraction regime, the far-field distribution model of the distortion facula was established. An objective function was established for coordinate calibration. The Zernike polynomial coefficient was solved using the conjugate gradient method. Finally, the position correction of the distorted facula was realized. Theoretical considerations and experiments demonstrated that this method improves the position precision of the beacon compared with the traditional method. The practicability of this method has also been verified by in-orbit optical communication experiment(HY-2 Satellite, China). The results of this research are likely to be useful for maintaining the laser link stability in inter-satellite optical communication systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] A. Belmonte, J.M. Kahn, Opt. Express 17 (2009) 12601–12611.
[2] D.M. Boroson, B.S. Robinson, Space Sci. Rev. 185 (2014) 115–128.
[3] S. Yu, Z. Ma, J. Ma, Opt. Express 23 (2015) 7263–7272.
[4] Y. Wang, D. Wang, J. Ma, IEEE Photonics J. 7 (2015) 1–10.
[5] P. Zhang, K. Qin, J. Huang, Opt. Commun. 344 (2015) 113–119.
[6] E. Segato, V.D. Deppo, S. Debei, G. Naletto, G. Gremonese, E. Flamini, Appl. Opt. 50 (2011) 2836–2845.
[7] Y. Yang, L. Tan, J. Ma, Appl. Opt. 48 (2009) 786–791.
[8] L. Tan, Y. Yang, J. Ma, J. Yu, Opt. Express 16 (2008) 13372–13380.
[9] H. Zhang, H. Zhao, Z.Zhao, Y. Zhuang, C. Fan, Opt. Express 27 (2019) 10495–10508.